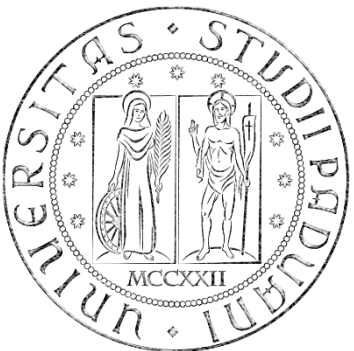


Biomedical Wearable Technologies
for Healthcare and Wellbeing

Setup the environment

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Giacomo Cappon



We need some tools in our belt

- Developing mobile apps requires some tools
- As programmers, we need to setup our **development environment** in order to be able to write code, compile it, test its behaviour, and deploy it to the final user machine (in this case, a phone).
- To do so, we need to:
 - Use some software to write the actual code
 - Choose a framework and the respective programming language
 - Have specific libraries in place to support the phone operating system
 - Have some tools to be able to work as a team
- This document will give an overview of the development environment we are going to use during this course and will tell what to do to prepare it.

The environment: Overview



IDE

(To write code,
compile, and test)



Flutter + Dart

(The framework and its
programming language)



Android Studio

(For Android
support)



XCode

(For iOS
support)



VCS

(For version
control and to
enable teamwork)

The environment: IDE

- The first component of the environment is the IDE (Integrated Development Environment).
- The IDE is where we actually will write the code: it is a text editor with some flavour (high-level functionalities).
- The IDE of choice in this course is Visual Studio Code (VS Code)



The environment: Framework and compiler

- The second component of the environment is, of course, the framework (and the programming language) we are going to use to develop mobile apps.
- We will use Flutter: a brand-new framework by Google based on the Dart programming language.
- Why Flutter? Because it allows us to **write a single code to create mobile app for either iOS or Android (but also desktop app for Windows, Mac, and Linux)**. This means that:
 - We will build one app that will look the same in both iOS and Android
 - We will not be constrained by the operating system (OS)
 - Developing time is halved

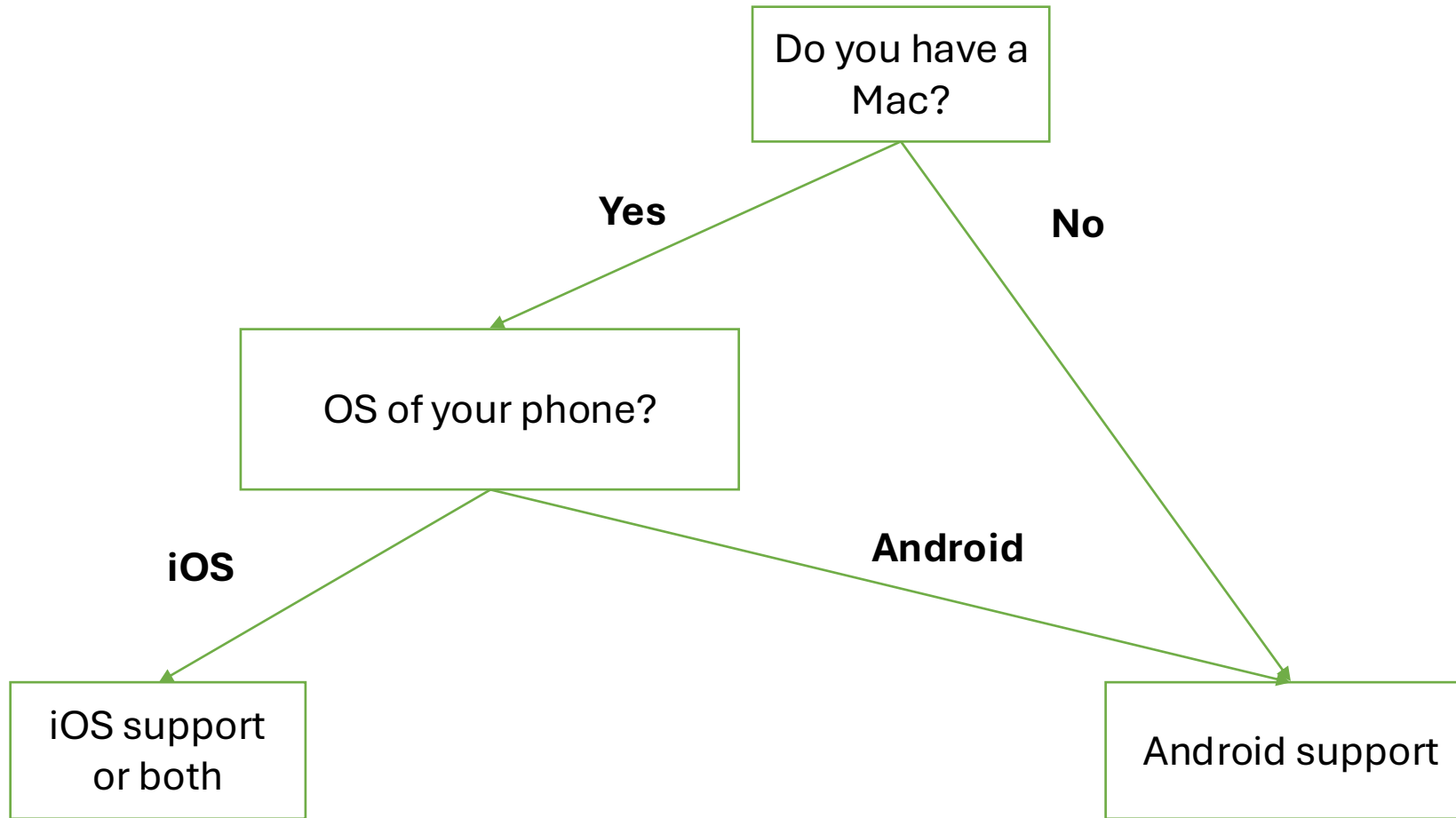


The environment: OS support

- The third component of the environment are the OS-specific (iOS or Android) libraries to install in order to let Flutter do its magic and compile.
- Both the iOS and the Android libraries will install the compiler for Flutter and a virtual phone simulator to allow you to test the mobile app without actually having a physical device.
- Since Flutter is OS-agnostic, you have a choice here:
 - Install iOS support
 - Install Android support
 - Install both
- A good way to choose the best option is: “Ok, I will have a virtual device, but at some point, I would like to deploy my app to an actual phone (my phone). What is the OS of my phone?”
- Note that if you have an iPhone but you do not have a Mac, you need to go for Android since XCode is not available for Windows



iOS or Android support?



The environment: VCS

- The final component of the environment is the Version Control System (VCS)
- As you will learn in the first lab lesson, the VCS is a software that allows to maintain and manage the various version of the code you are going to write, and it will be fundamental to work as a team.
- In this course, we will use GIT, the most famous VCS.



Instructions

➤ Mac Users need to install the following:

1. XCode
2. Rosetta (if you have an Apple Silicon)
3. VS Code
4. Flutter and Cocoapods
5. Android Studio (if Android support is necessary)

Slides 10-14



➤ Windows Users need to install the following:

1. Git
2. VS Code
3. Flutter
4. Android Studio

Slides 15-18



(Mac) 1. XCode

1. Open the App Store, search “XCode” and install it



(Mac) 2. Rosetta

- When installing on an Apple Silicon Mac, install the Rosetta translation environment. Some components require the Rosetta.

1. Open a Terminal and run

```
>> sudo softwareupdate --install-rosetta --agree-to-license
```

```
cappe@Giacomos-MacBook-Pro ~ % sudo softwareupdate --install-rosetta --agree-to-license
```



(Mac) 3. VS Code

1. Install VS Code

- Go to > <https://code.visualstudio.com/>
- Download and install VS Code

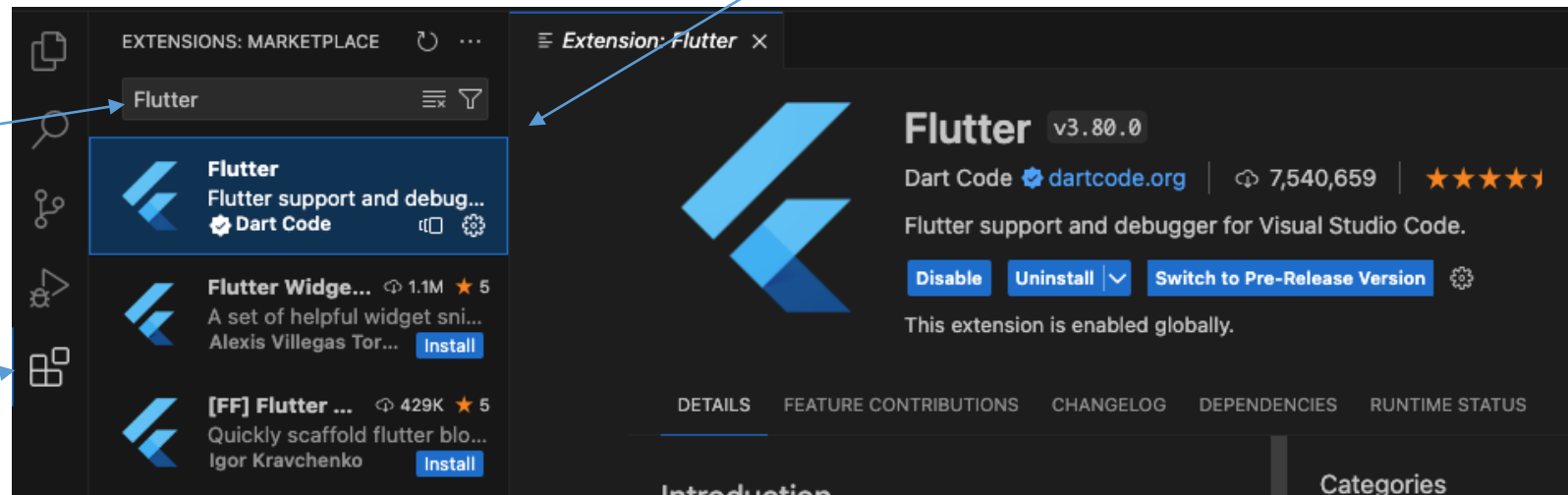
2. Add the Flutter extension for VS Code

- Open VS Code
- Go to the Extension tab
- Type 'Flutter' in the search bar
- Install the first entry of the search results



Search bar

Flutter extension
to be installed



'Extension' tab

(Mac) 4. Flutter

1. Go to <https://docs.flutter.dev/install/quick#install>
2. Follow the instructions from section "Install and set up Flutter" to the end



(Mac) 5. iOS support

1. Go to <https://docs.flutter.dev/platform-integration/ios/setup>
2. Follow the instructions from section "Set up iOS tooling" to the end



(Mac) 6. Android Studio

1. Go to <https://developer.android.com/studio>, download Android Studio and install it
2. Go to <https://docs.flutter.dev/platform-integration/android/setup#set-up-tooling>
3. Follow the instructions from section "Set up Android tooling" to the end



(Windows) 1. Git

1. Simply go to <https://gitforwindows.org/>, download Git SCM and install it



(Windows) 2. VS Code

1. Install VS Code

- Go to > <https://code.visualstudio.com/>
- Download and install VS Code

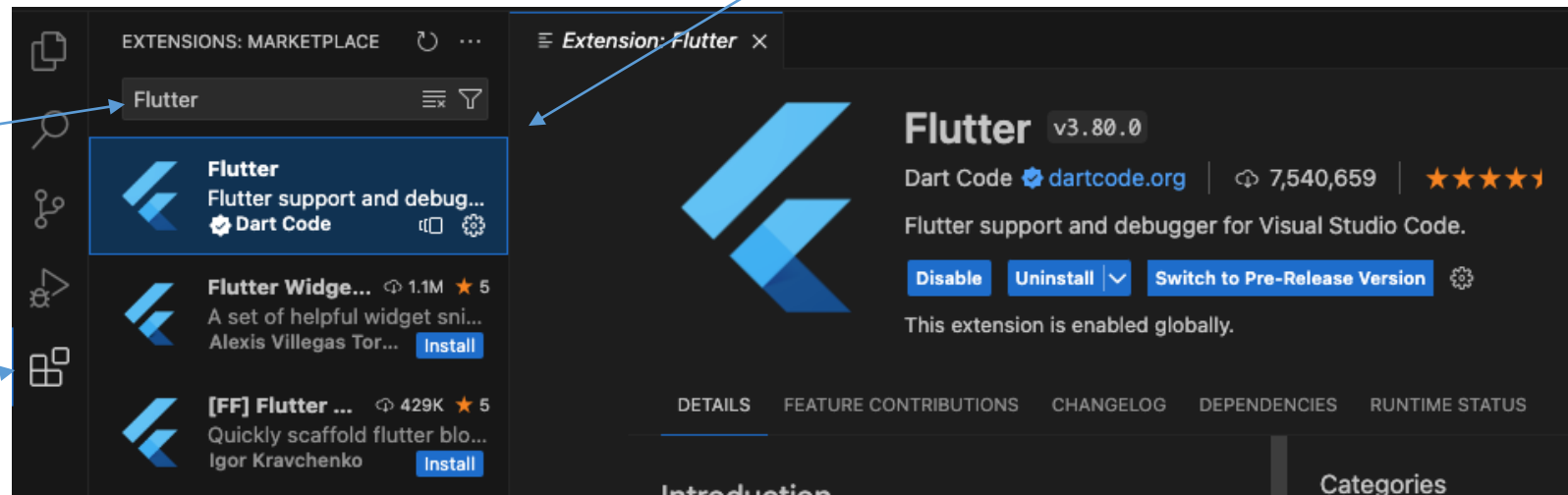
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Search bar

Flutter extension
to be installed



'Extension' tab

(Windows) 3. Flutter

1. Go to <https://docs.flutter.dev/install/quick#install>
2. Follow the instructions of section "Install and set up Flutter" to the end



(Windows) 4. Android Studio

1. Go to <https://developer.android.com/studio>, download Android Studio and install it
2. Go to <https://docs.flutter.dev/platform-integration/android/setup#set-up-tooling>
3. Follow the instructions from section "Set up Android tooling" to the end

